

Number Sense

Integers · BEDMAS · Exponents · Fractions · Roots · Percent · Ratios

MTH1W · Homework — Independent Practice

Name: _____

Date: _____

Goal: Practise integers, exponent laws, fractions, roots, percents and ratios independently.

Show all your work. Levels: L1 fluency, L2 application, L3 thinking. Full worked solutions are in the companion Answer Key.

1. Evaluate: $(-12) - (-7) + (-3)$. **[L1]**

2. Evaluate using BEDMAS: $40 - 6 \times 3 + 2^2$. **[L1]**

3. Simplify (leave in exponent form): $7^3 \times 7^2$. **[L1]**

4. Simplify, $m \neq 0$: $m^5 \div m^2$. **[L1]**

5. Evaluate in lowest terms: $5/8 + 1/6$. **[L1]**

6. Write 0.24 as a fraction in lowest terms and as a percent. **[L1]**

7. Is $\sqrt{81}$ rational or irrational? Give its value. **[L1]**

8. Evaluate in lowest terms: $2/5 \times 3/4 \div 1/2$. **[L2]**

9. Simplify to a single power and its value: $(3^2)^3 \div 3^4$. **[L2]**
10. A \$60 pair of shoes is marked up 25%. Find the new price. **[L2]**
11. Order from least to greatest: -0.4, $-\frac{1}{3}$, -0.5, $\frac{1}{5}$. **[L2]**
12. On a map, 1 cm represents 25 km. Two towns are 7 cm apart on the map. Find the real distance. **[L2]**
13. 3 kg of apples cost \$8.40. At the same rate, find the cost of 5 kg. **[L2]**
14. Between which two consecutive whole numbers does $\sqrt{30}$ lie? Justify. **[L2]**

15. After a 20% discount, a coat costs \$96. Work backwards to find the original price, and check. **[L3]**
16. Maya wrote $(2^3)^2 = 2^5$. Explain her error and give the correct value. **[L3]**
17. A 500 mL drink is 40% juice. How much pure juice must be added so the drink is 50% juice? **[L3]**
18. A cyclist rides 36 km at 12 km/h, rests, then rides 20 km at 10 km/h. Find the total riding time and the average speed over the distance ridden. **[L3]**